Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/08/2008
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Dose rate effects on the mechanisms of space radiation induced delayed genomic instability		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Morgan, William (University of Maryland)		
Grant/Contract No.:	NNX07AT42G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Those brave individuals traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to a greater risk for developing cancer later. Since there is currently no way to protect astronauts from space radiations, it is important that the biological consequences of exposures to these radiations are understood. Using proton and iron ion radiations we will investigate the delayed effects of radiation exposure based on the type of radiation and the rate at which the radiation dose is delivered. Specifically, we will investigate those deleterious effects observed in the progeny of irradiated cells multiple cell generations after the initial exposure. These deleterious effects are grouped under the term of radiation-induced genomic instability and are thought to be an early step in the carcinogenic process. The primary objective of this proposal is to evaluate the differences in dose rate effects and radiation quality among x-rays, protons, and iron ions in the induction of delayed genomic instability and bystander effects. We will accomplish this by measuring cell survival, gene mutations/deletions and rearrangements, and other forms of chromosome damage. Further this project will explore the possibility that DNA methylation is involved in the		

	mechanisms underlying radiation induced genomic instability.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the research and technology questions that this proposal addresses include the following: 1) What are the probabilities for increased carcinogenesis from space radiation as a function of NASA's operational parameters (age at exposure, age, latency, gender, tissue, mission, radiation quality, dose rate, and exposure protraction)? 2) How can effective biomarkers of carcinogenic risk from space radiation be developed and validated? 3) What is the likelihood that DNA damage will increase the risk of carcinogenesis? We hypothesize that dose rate and radiation quality affect the induction of delayed genomic instability and aberrant DNA methylation (thought to be cancer precursors) following exposure to space radiations. Specific Aim 1 will test the hypothesis that dose rate and radiation quality affect the induction of delayed genomic instability following exposure to radiations commonly encountered in space, specifically iron ions and protons. Using a panel of assays (e.g. chromosomal instability, micronuclei (MN) formation, delayed mutation, and reproductive integrity) we will compare the biological effectiveness of dose rate effects and radiation quality among protons, low LET x-ray radiation and high LET heavy ion radiation for the induction of delayed genomic instability. We hypothesize that altered DNA methylation is an epigenetic effect of exposure to space radiations that has a mechanistic role in genomic instability and is a measure of carcinogenic risk. We will provide a greater understanding of the potential relationship between radiation-induced genomic changes and epigenetic factors based on radiation quality, dose and dose rate. We will develop and validate high throughput biomarkers for genomic instability endpoints and DNA methylation. These hypotheses will be tested in Aims 1 and 3. In Specific Aim 3 we will evaluate DNA methylation globally and at specific loci within the genome before and at immediate and delayed times post-irradiation following exposure to high and low dose rate protons at high and low energies, iron ions and x-rays. Among the loci evaluated for changes in methylation in this study are tumor suppressor genes for whom inappropriate promoter methylation leads to gene silencing. This promoter hypermethylation and gene silencing has been linked to a number of specific cancers, and is thought to be a biomarker for increased carcinogenic risk. These methylation assays are PCR based and will provide a high-throughput assay for the assessment of carcinogenic risk.
Task Progress:	During the first year of this project, a technician was hired and trained to work 100% time on this project. During this time, I also began developing a panel of DNA methylation assays. Currently in place are two assays to evaluate global DNA methylation. The first technique is a methylation sensitive arbitrarily priming (MSAP) PCR that detects qualitative changes in genomic DNA methylation. We have already applied this technique to detect differences in DNA methylation between the stable parental Chinese hamster ovary-human hybrid cell line GM10115 and the LS12 genomically unstable GM10115 clone. The LS12 clone was clonally expanded from a population of cells exposed to 10 Gy of x-rays and determine to be genomically unstable by cytogenetic analysis (fluorescence in situ hybridization for human chromosome 4). The second assay evaluates the methylation status of long interspersed nucleotide elements (LINE-1) and Alu elements within the genome. We have validated this technique by detecting differences in methylation between RKO36 cells treated with the DNA methylation inhibitor 5-aza-2'deoxycytidine (DAC) and vehicle controls. The DAC induced decrease in DNA methylation that we observed in these experiments is similar to that reported in the literature (20.7% +/- 1.1% and 4.4% +/- 0.2% for LINE-1 and Alu, respectively). Together, the MSAP PCR and repeat element assays provide a measure of change in genome-wide methylation. Assays to evaluate methylation at promoter regions of genes involved in tumor suppression and cell cycle regulation are being optimized. Gene silencing of these loci has been linked to carcinogenesis or an increased risk for cancer. Together with our genomic instability assays and our global and specific locus DNA methylation assays, we will be able to determine whether there is a link among radiation induced genomic instability, changes in DNA methylation, and radiation quality and dose rate. We received one hour of proton beam time during the Spring 2008 campaign by piggybacking with Dr. Sutherland's team. This hour allowed us to do some preliminary experiments using the human colon carcinoma cell line RKO and the normal human fibroblast cell line AGO1522. Cells were irradiated using 150 MeV protons (0-200 cGy, ~33 cGy/min) and then evaluated for clonogenic cell survival and DNA damage as measured by the micronucleus assay (MN). We were awarded six hours of our own time during the Summer campaign. We used this time to initiate replicate lower dose rate experiments using the 150 MeV protons (~ 2 cGy/min), again evaluating survival and MN induction immediately following irradiation. Cell stocks from both of these runs will be grown and evaluated for radiation induced genomic instability and effects on DNA methylation.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Abstracts for Journals and Proceedings	Aypar U, Malkiel LN, Baulch, JE. "Effects of high and low LET irradiation on DNA methylation and genomic instability." 54th Annual Radiation Research Society Meeting, Boston MA, September 21-24, 2008. Radiat Res, in press, July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations in Radiation Induced Genomic Instability." 54th Annual Radiation Research Society Meeting, Boston, MA, September 21-24, 2008. Radiat Res, in press, July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations in Radiation Induced Genomic Instability." 19th Annual NASA Space Radiation Health Investigators'Workshop, Philadelphia, PA, June 30-July 2, 2008. Workshop Abstracts, 19th Annual NASA Space Radiation Health Investigators' Workshop, June 2008. , Jun-2008
Abstracts for Journals and Proceedings	Aypar U, Malkiel LN, Baulch JE. "Effects of high and low LET irradiation on DNA methylation and genomic instability." 39th Annual Environmental Mutagen Society Meeting, Rio Grande, Puerto Rico, October 18-22, 2008. Environ Mol Mutagenesis, in press, July 2008. , Jul-2008

Abstracts for Journals and Proceedings	Aypar U, Morgan WF, Baulch JE. "Epigenetic Alterations In High And Low LET Radiation Induced Genomic Instability." 39th Annual Environmental Mutagen Society Meeting, Rio Grande, Puerto Rico, October 18-22, 2008. Environ Mol Mutagenesis, in press, July 2008. , Jul-2008